Unit-5 Client Server and Web-based Programming Technology

Er. Bhisma Raj Joshi
Department of Computer Engineering
[NAST]

@Er BHISMARA

OBJECTIVES

- Standalone Programming Concept(Adv and Limitations)
- Client Server Programming(Advantages and Limitations)
- Differences between Standalone & Client server Programming
- Client Server Programming Architecture
- Web-bases Programming and it’s Architecture
- Advantages and Disadvantages
- Some important web-based programming and applications
- Introduction to mobile computing languages and their application

@Er BHISMARA

EXAM POINT OF VIEW

8HR

2 Long Questions 7+8 =15

1 Short Note 1*5 = 5

@Er BHISMARA

Standalone Programming Concept

- Standalone system is a type of system that does not have any dependencies or is not connected to any other system.
- Standalone program is a computer program that does not load any external modules, library function or any other files.
- It does not require a database, a network connection for operation and is only dependent on the local computer resources.
- This concept is not widely used now days because they often lack the connectivity, flexibility, and scalability that modern interconnected systems and networks provide.

@Er BHISMARA

- Standalone programming refers to creating software applications that can operate independently on a computer without requiring network connectivity or other external resources.
- Common examples include desktop software like Microsoft Word, Adobe Photoshop, Notepad, Calculator and various games that don't require an internet connection.

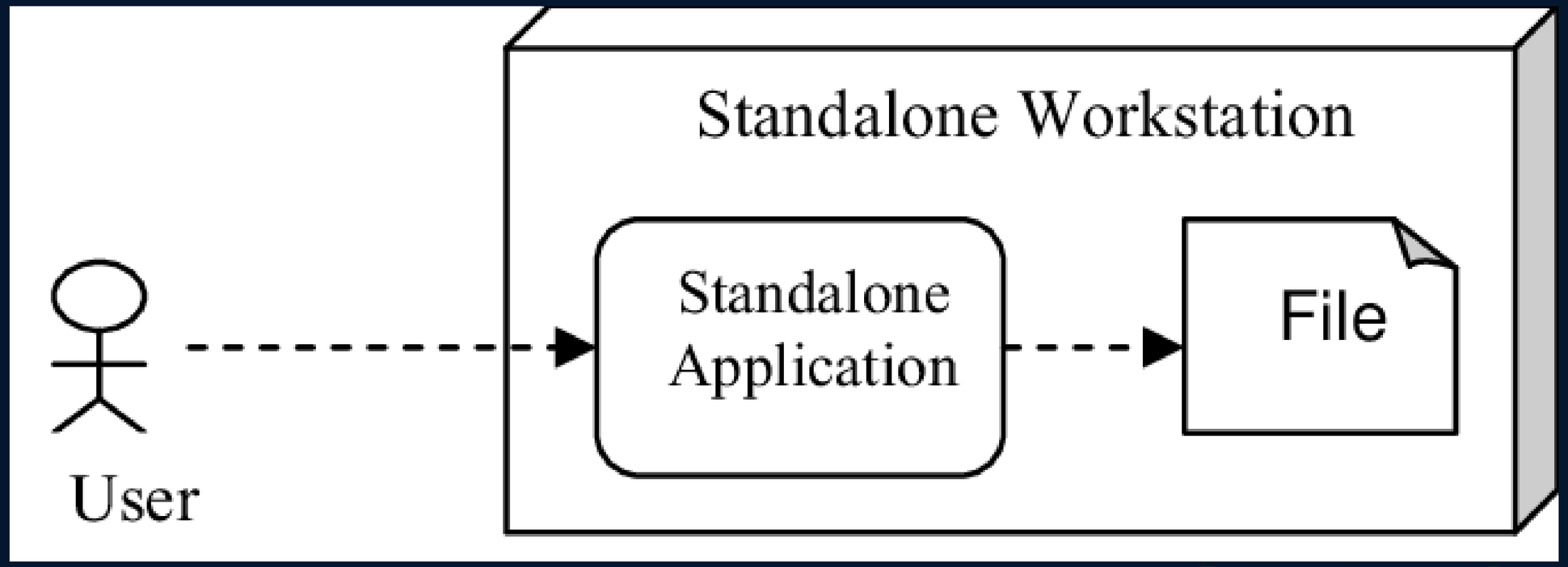


Fig:- Stand-alone system

@Er BHISMARA

Advantages of Standalone Programs

1. Independence:- Standalone programs do not rely on external systems, networks, or internet connectivity.
2. Simplicity:- These programs are typically simpler to install, manage, and use since they do not require complex integration or additional setup.
3. Performance:- Since standalone programs run locally on the device, they often deliver faster performance.
4. Security:- By operating in isolation, standalone programs reduce the risk of external cyberattacks, data breaches, or unauthorized access.

@Er BHISMARA

Dis-advantages of Standalone Programs

1. Lack of Connectivity:- Standalone programs do not support real-time data sharing or communication with other systems.
2. Limited Scalability:- Expanding the functionality or user base of standalone programs often requires significant manual effort or complete redevelopment.
3. Maintenance Challenges:- Updates and bug fixes must be applied individually to each system, which can be time-consuming and inefficient for larger deployments.
4. Data Isolation:- Data created or saved in standalone programs usually stays on that one computer and cannot be easily shared or accessed by other systems.

@Er BHISMARA

Client Server Programming Concept

- The Client-server system or programming is a distributed application structure where the overall task are divided between two types of system which are called client and server.

- Since client is a device or program that requests services or resources(e.g. a web browser). Client always sends requests to the server and waits for the response.

- Server is a device or program that provides services or resources back to the clients(e.g. a web server). It always listens for clients requests, process them and send responses.

@Er BHISMARA

- Here are few examples of client server system, online gaming(gaming console like PUBG, COD are clients and AWS, Google clouds are servers) similarly database system(MySQL workbench act as a client and MySQL as server) Email system(email apps like Gmail, outlook are clients and SMTP, IMAP are mail servers).

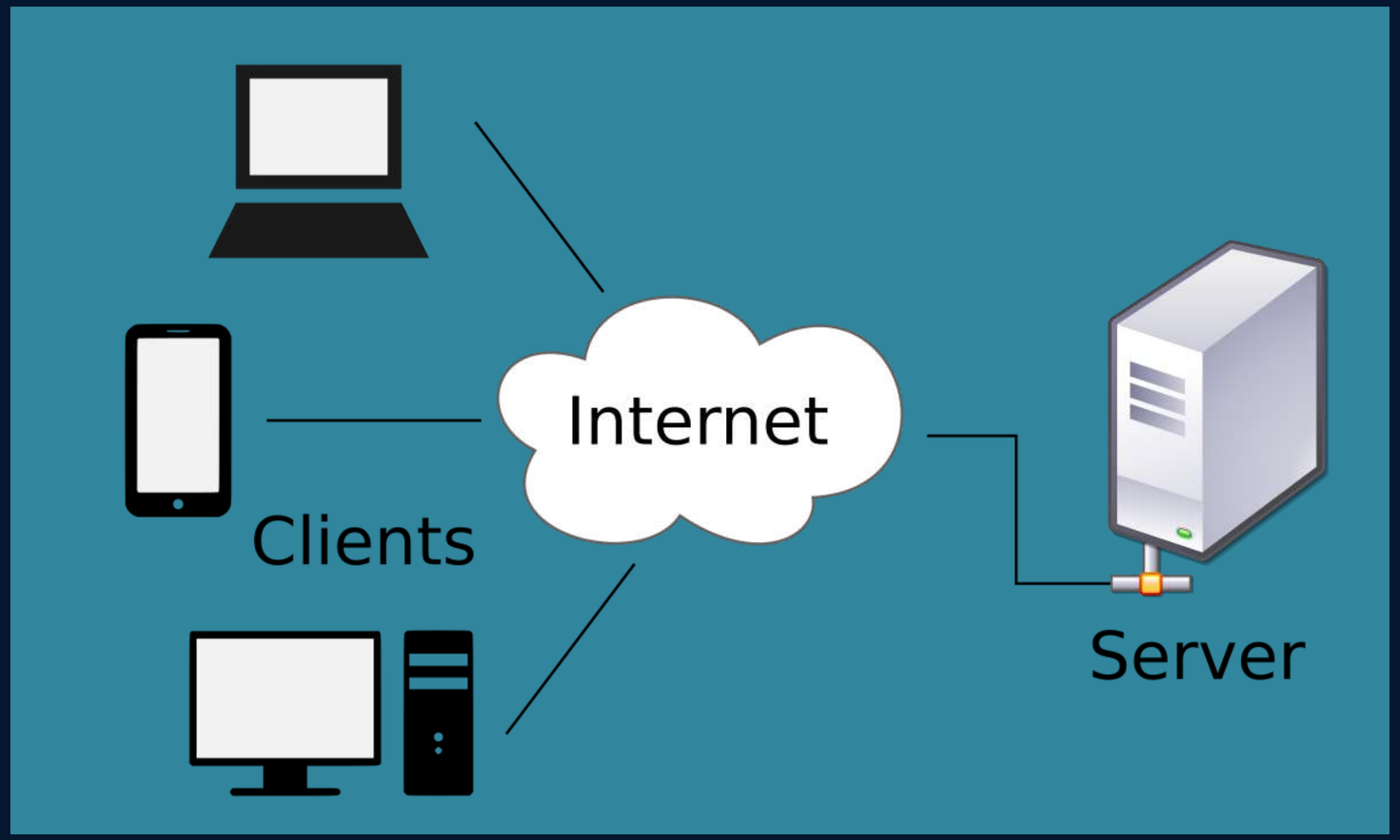


Fig:- Clients-server system

@Er BHISMARA

Advantages of Client-Server system

1. Scalability: Servers can handle multiple clients simultaneously, making it easy to scale the system as the number of users grows.
2. Centralization: All the files and records are stored in the centralized server. So it is easier to manage the files and access them.
3. Resource Sharing: Efficiently shares resources, such as data and information among multiple clients, leading to cost savings.
4. Improved Data Security: Centralized control allows for better implementation of security measures, data backups, and access control, safeguarding sensitive information.
5. Easy Backup: Since all files are stored in the single location so it is easier to backup them in secondary storage device.

@Er BHISMARA

Limitations or dis-advantages of CS

1.Single Point of Failure: If the server fails, all clients relying on it can lose access to the services, causing interruptions.

2. Network Dependency: Client-server systems rely heavily on a stable and reliable network connection. Poor network performance can lead to slow response times or disconnections.

3. Scalability Challenges: While client-server systems can scale, the server may become a bottleneck if not properly managed, requiring additional hardware and resources to handle increased load.

4. Maintenance Costs: Centralized management and maintenance of the server require skilled person and continuous monitoring, which can increase operational costs.

@Er BHISMARA

ASSIGNMENT

"WHAT ARE THE KEY DIFFERENCES BETWEEN STANDALONE APPLICATIONS AND CLIENT-SERVER APPLICATIONS IN TERMS OF DEPLOYMENT, MAINTENANCE, PERFORMANCE, NETWORK DEPENDENCY AND SECURITY?

@Er BHISMARA

Client-Server Programming Architecture

- Client-server architecture is a fundamental concept of modern system design, where the network infrastructure is structured to include multiple clients and a central server.

- In this model, clients are devices or programs that make requests for services or resources, while the server is a powerful machine or software that fulfils these requests.

- Communication between clients and the server follows a request-response protocol, such as HTTP/HTTPS for web services or SQL for database queries.

@Er BHISMARA

- Developers commonly use one of the following client-server architecture while designing the system.

1. 2-Tier Architecture

- A system design in which the user interface (client) and the data layer (server) are separated into two distinct tiers. The client directly communicates with the server, without any intermediary layers.

- With two-tier client/server architectures, the user system interface is usually located in the user’s desktop environment and the database management services are usually in a server that is a more powerful machine that services many clients.

@Er BHISMARA

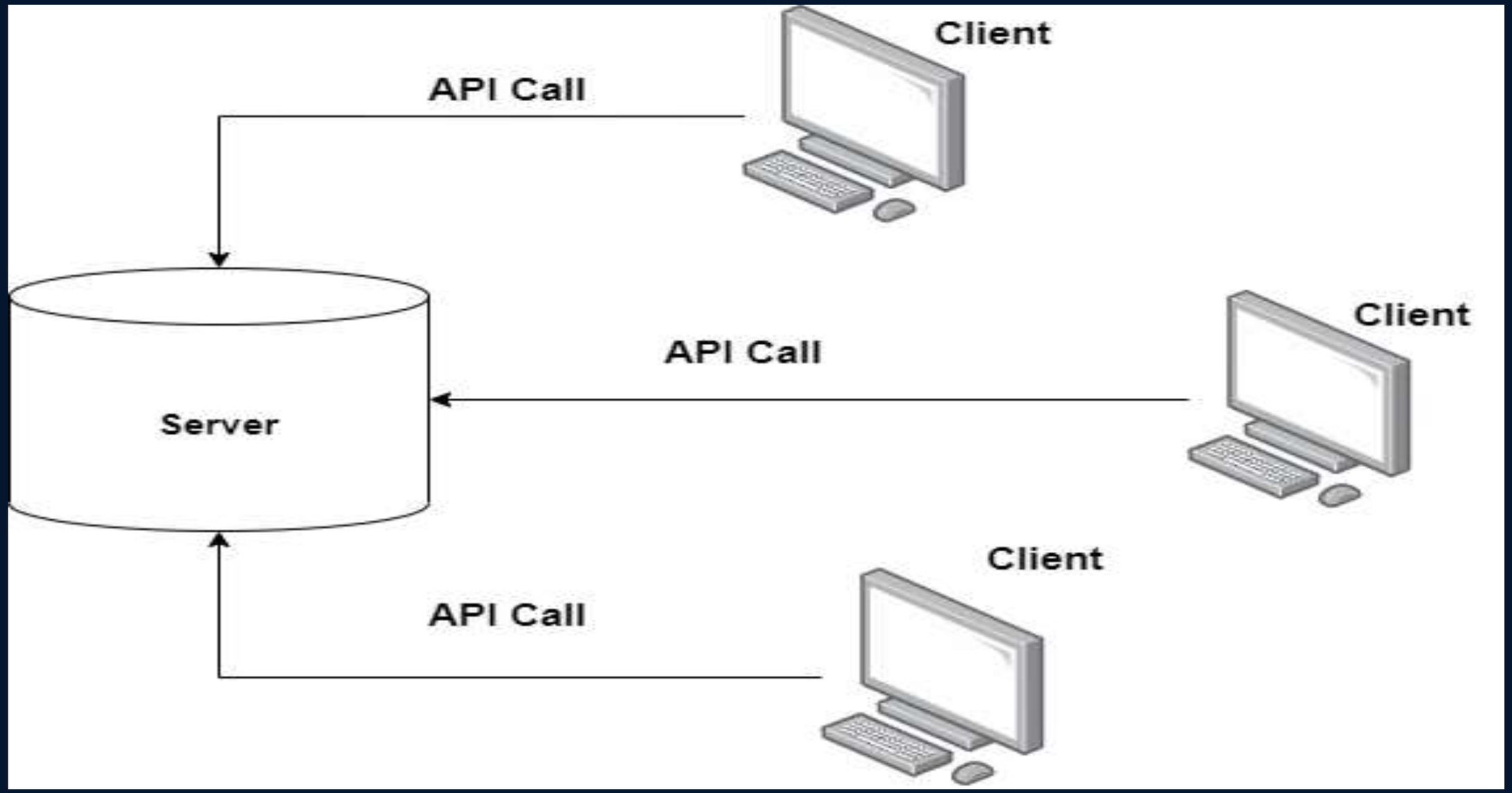


Fig:- 2 Tier client server architecture

@Er BHISMARA

Workflow:

1. Client Request: The client application sends a request to the server, typically for data retrieval or updates.
2. Server Processing: The server processes the request, performs the necessary operations on the database, and prepares the response.
3. Server Response: The server sends the response back to the client, which then displays the information to the user.

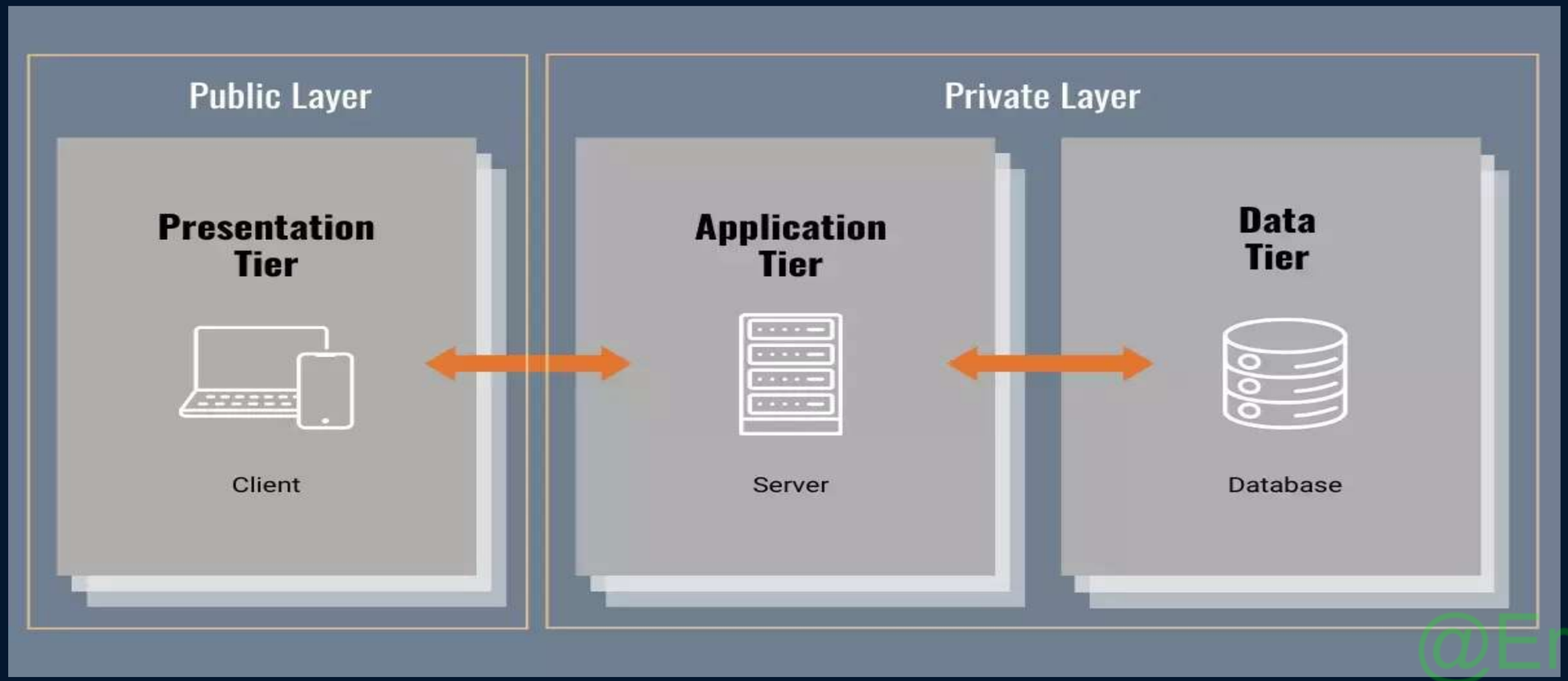
Key Features:

1. Direct Communication: The client directly interacts with the server, leading to simple and efficient communication.
2. Simplified Development: Easier to develop and maintain due to the straightforward interaction between client and server.
3. Lower Latency: Reduced latency as there are fewer intermediary layers compared to multi-tier architectures.

@Er BHISMARA

2. 3-Tier Architecture

- A system design that separates an application into three distinct layers or tiers: the presentation layer, the application layer, and the data layer. Each tier is responsible for specific functions and interacts with the other tiers in a structured manner.



@Er BHISMARA

Key Components:

1.Presentation Layer (Client): It handles the user interface and presentation logic. It is responsible for displaying data to the user and collecting user input. Example: Web browsers, mobile apps, desktop applications.

2.Application Layer (Middle Tier/Logic Layer): It contains the business logic and processes user requests. It acts as an intermediary between the presentation layer and the data layer. Example: Web servers, application servers, APIs.

3.Data Layer (Database Server): It’s main function is to manages data storage, retrieval, and database management. It is responsible for maintaining data integrity and consistency. Example: Database servers like MySQL, Oracle, SQL Server.

@Er BHISMARA

Workflow

Client Requests: The user interacts with the client (presentation tier) through a web browser or an application. User actions, such as clicking a button or submitting a form, send a request to the application tier.

Business Logic Processing: The application tier (business logic tier) receives the request and processes it based on the rules and logic. It may involve computations, decision-making, and validation of user input. If needed, the application tier will interact with the data tier to retrieve or store data.

Data Management: The data tier (database tier) handles requests from the application tier to access or modify data. It ensures data integrity, consistency, and security while processing these requests.

@Er BHISMARA

Response Generation: Once the application tier has processed the request, it generates a response. The response is sent back to the presentation tier for display to the user.

Client Response: The presentation tier receives the response and updates the user interface accordingly. The user can now view the updated information or proceed with further actions.

@Er BHISMARA

ASSIGNMENT

WRITE DOWN THE MAJOR DIFFERENCES BETWEEN 2-TIER AND 3-TIER CLIENT SERVER ARCHITECTURE.

@Er BHISMARA

Web based Programming

- Web-based programming commonly known as web development, refers to the process of developing different websites and web applications that are designed to run on web servers and are accessible to users via web browsers over the internet.

- It encompasses a wide range of technologies, frameworks, and tools that allow developers to create interactive, dynamic, and user-friendly web applications.

- This process is divided into two parts that are

i) Front-end Development
ii) Backend Development

@Er BHISMARA

1. Front-End Development

- Front-end development focuses on the client-side(GUI) development. It is the part of the web application where users interact. Here are some core technologies or programming languages used for front-end development:

•HTML (Hyper-Text Markup Language): The foundation of web pages, HTML provides the structure and layout.

•CSS (Cascading Style Sheets): This is used to style the layout of the HTML elements. CSS controls the visual presentation and design.

•JavaScript: A versatile language that enables dynamic and interactive elements on web pages. JavaScript can update and change both HTML and CSS.

@Er BHISMARA

ii) Back-End Development

- Back-end development involves server-side programming that handles the logic, database interactions, user authentication, and more. Key components include:

Server: The hardware and software that serve the content to the client. Popular server-side languages include Python, Ruby, PHP, Java, and (JavaScript).

Database: Stores and retrieves data for the web application. Common databases are MySQL, PostgreSQL, MongoDB, and SQLite.

APIs (Application Programming Interfaces): Allows different software systems to communicate. REST and GraphQL are widely used for creating APIs.

@Er BHISMARA

Advantages of web-based programming

1. Highly Accessible:- Web applications and websites are accessible from anywhere with an internet connection and a web browser.
2. Cross-Platform Compatibility: Web-based programs run on any platform Windows, macOS, Linux, Android, iOS without the need for platform-specific development.
3. Easy for updates and Maintenance:- Easy to update bugs or security patches can be quickly resolved in a single location(server).
4. Cost Effectiveness:- A single application can work across multiple platform which reduce the development cost.
5. Scalability:- Web-based applications hosted on cloud platforms can scale up or down easily based on traffic demand.

@Er BHISMARA

Dis-advantages of web-based programming

1. Dependency on Internet:- The web-based programs are dependent on the internet connectivity, so it may be slower response if the internet connection is slower.
2. Security Concerns:- Being internet-based, web applications are vulnerable to attacks like hacking, phishing, DDoS attacks, and malware.
3. Browser Issues:- Users with outdated browsers may face degraded performance or usability issues.
4. Dependency on Servers:- High traffic loads can overwhelm servers if not adequately prepared, if the server down the application is also down.

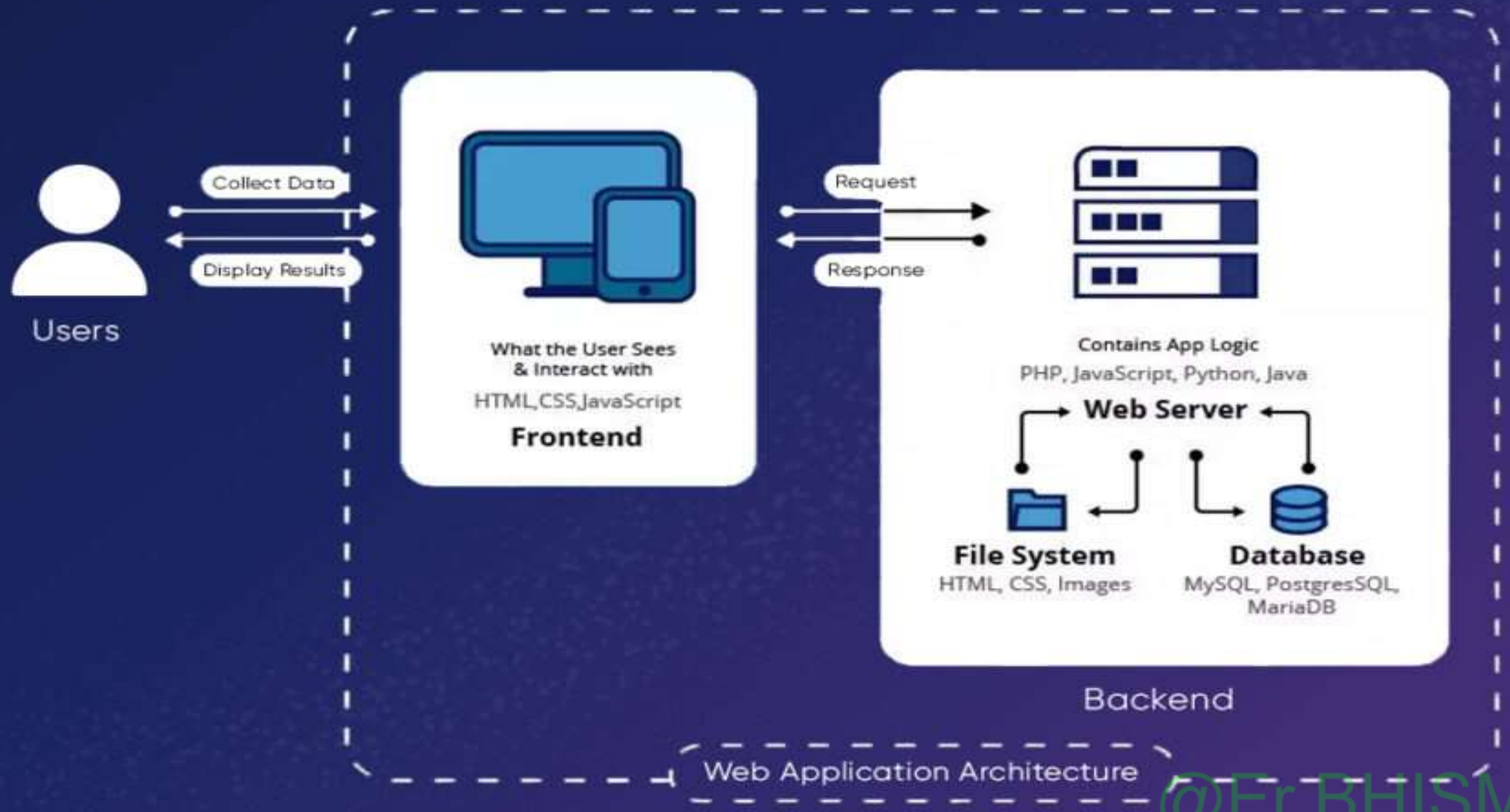
@Er BHISMARA

Architecture of web-based programming

- A website is a static collection of web pages, that contain different contents like images, text, audio, video etc and are accessed through browser.
- Whereas web application is an interactive, dynamic software program that runs on a web server and requires user input.
- So web-based architecture is a mechanism that gives us a clarification that how the connection is established between the client and the server.
- It determines how the components in an application communicate with each other. It doesn’t matter what is the size and the complexity level of the application is, they all follow the same principle.

@EFBIISMARA

WEB APPLICATION ARCHITECTURE



@Er BHISMARA

- In technical terms, when a user makes a request on a website, various components of the applications, user interfaces, middleware systems, databases, servers, and the browser interact with each other.

- So in one sentence web-based Architecture is a framework that ties up this relation of different component together and maintains the interaction between these components.

@Er BHISMARA

Some important web-based programming and applications

- There are several key web-based programming languages and applications that have transformed the way we interact with the web. Here's a quick overview of some of the most important ones:

Programming Languages

HTML/CSS: The backbone of web development, HTML provides the structure of web pages, while CSS is used for styling and layout.

JavaScript: A versatile language that allows for dynamic, interactive features on websites and web applications.

@Er BHISMARA

Python: Often used for server-side web development due to its simplicity and readability.

Ruby: Known for its elegant syntax, used primarily with the Ruby on Rails framework.

PHP: A server-side scripting language designed for web development but also used as a general-purpose programming language.

@Er BHISMARA

Frameworks and Libraries

React.js: A JavaScript library for building user interfaces, particularly single-page applications.
Angular: A TypeScript-based open-source web application framework led by the Angular Team at Google.
Django and Flask: A high-level Python web framework that encourages rapid development and clean, pragmatic design.
ASP.NET: A web framework for building dynamic websites and web applications. It includes tools and libraries for building web APIs, web pages, and real-time communication.

@Er BHISMARA

Applications

- Here are some key examples of applications that are used in web technology.

Content Management Systems (CMS): WordPress, Joomla, and Drupal are popular CMS platforms that allow users to create, manage, and modify content on a website without needing specialized technical knowledge.

E-commerce Platforms: Shopify, Alibaba, and Amazon enable businesses to set up online stores and manage their sales and inventory.

Social Networking Sites: Facebook, Twitter, and LinkedIn are examples of web applications that connect users and facilitate communication and networking.

@Er BHISMARA

Email Services: Gmail, Outlook, and Yahoo! Mail are web-based email services that allow users to send and receive emails from anywhere with an internet connection.

Online Collaboration Tools: Google Workspace (formerly G Suite), Microsoft Office 365, and Slack are platforms that provide various tools for team collaboration and productivity.

These tools and technologies form the foundation of modern web development and power many of the applications and services we use every day.

@Er BHISMARA

Mobile Computing Language

- Mobile computing is the use of portable computing devices (such as smartphones, tablets, laptops, and wearables) to access, process, and store information without being confined to a single location.
- These devices are often connected to the internet, allowing users to access cloud-based services, applications, and data from anywhere.
- The term "mobile computing language" can refer to programming languages and tools used to develop applications for mobile devices such as smartphones, tablets, and wearables. Popular programming languages for mobile computing include:

@Er BHISMARA

Java: A robust and widely-used language for Android app development, known for its portability and ease of use.

Kotlin: A newer, simpler language that works well with Java and separately. You can use both languages in the same project without issues.

Swift: Swift is a fast, safe, and modern programming language designed by Apple for iOS, macOS, watchOS, and tvOS development. It features a clean and concise syntax, making it easier to read and write code. Swift is the preferred language for developing apps within the Apple ecosystem due to its performance and safety advantages.

@Er BHISMARA

JavaScript (JS): JavaScript is a versatile language primarily used for creating interactive web applications. It can also be used with frameworks like React Native to develop cross-platform mobile apps. It's essential for adding dynamic features to websites and applications.

Dart: Dart is a programming language developed by Google, used primarily with the Flutter framework for building high-performance cross-platform mobile applications. It features a clean syntax and allows developers to write code that runs on both iOS and Android. Dart’s performance and flexibility make it ideal for crafting modern mobile apps.

@Er BHISMARA

C#: It is a versatile language developed by Microsoft, used to build cross-platform mobile applications. It allows developers to write code that can run on iOS, Android, and Windows devices. C# is known for its strong typing and object-oriented features.

Python: Python is a powerful and easy-to-read language, often used for web development, data analysis, and automation. Although less common for mobile apps, it can be used with frameworks like Kivy to create mobile applications. Python's simplicity and readability make it a favourite among developers.

@Er BHISMARA

ASSIGNMENT

WHY MOBILE COMPUTING LANGUAGE IS POPULAR NOWADAYS? . WHICH WEB-BASED LANGUAGE WOULD YOU PREFER FOR DEVELOPING A WEB APPLICATION? EXPLAIN WITH REASONS

@Er BHISMARA